

Note: Write your answers clearly in the blanks as per the unit given.

The voltage and currents are in rms for AC circuits.

Q1-7 each correct blank carries 1 Mark and Q8-Q20 each correct blank carries 2 Marks.

Name: _____

ID No: _____

Sec. No. _____

- Q1. A MOSFET is a current-controlled device. _____ (True/False)
- Q2. Mobility of electrons is higher than mobility of holes. _____ (True/False)

- Q3. A BJT operates in the cut-off mode when the emitter-base junction is _____ biased, and the collector-base junction is _____ biased.

- Q4. The time constant for a first order series R-L ($R = 0.5 \Omega$ and $L = 2 \text{ mH}$) circuit is _____ s.

- Q5. In Fig.Q5, the power supplied by element P_4 is _____ W, and power absorbed by element P_3 is _____ W.

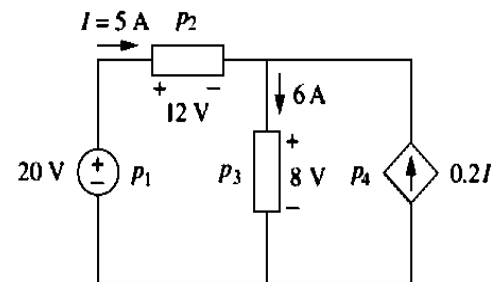


Fig.Q5

- Q6. Three light bulbs are connected to a 9 V battery as shown in Fig.Q6. The total current supplied by the battery is _____ A, the current through 20 W bulb is _____ A and the resistance of 10 W bulb is _____ Ω .

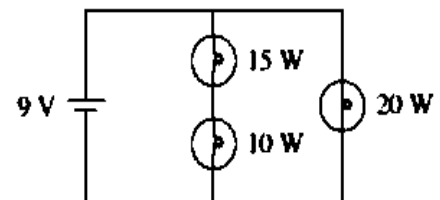


Fig.Q6

- Q7. In a series RLC circuit, at resonance the impedance of the circuit is _____. (maximum/minimum)

- Q8. In the circuit shown in Fig.Q8, value of V_x is _____ V.

- Q9. A series RLC circuit with $R = 10\Omega$, $X_L = 20\Omega$ and $X_C = 20\Omega$ is excited by a sinusoidal source of voltage 200V. The voltage across inductance is given by _____ V.

- Q10. A three phase, 400V (line) delta connected load draws a line current of 2.7A at 0.8pf lagging. The phase current is _____ A and the total reactive power drawn by the load is _____ VAR.

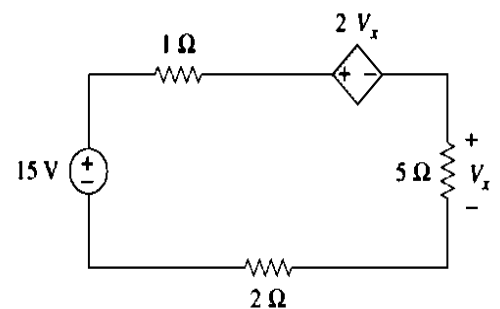


Fig.Q8

- Q11. A series RLC circuit excited by a 10V sinusoidal source resonates at a frequency of 50Hz. If the bandwidth is 31.4 rad/s then the quality factor at resonance is _____.
- Q12. A length of copper wire with a cross sectional area of 10^{-6} m^2 carries a current of 0.4 A. The conductivity of copper at 293K (20°C) is $5.78 \times 10^7 \text{ mho/m}$ and the free electron concentration is $8.43 \times 10^{28} \text{ m}^{-3}$. The drift speed of free electrons is _____ m/s and the electric field strength is _____ V/m.

Q13. For the circuit shown in Fig Q13, if $i > 0$ then the silicon BJT is operating in _____ mode.

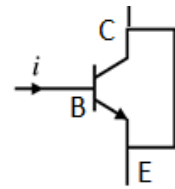


Fig.Q13

Q14. The maximum possible speed of a 50 Hz synchronous motor, is _____ rpm.

Q15. In a MOSFET the resistance between Gate and Source is very _____.(low/high)

Q16. An n-channel depletion MOSFET has $I_{DSS} = 10 \text{ mA}$, $V_p = -2 \text{ V}$ and $V_{GS} = 1 \text{ V}$. For small values of V_{DS} , the value of drain-source resistance (r_{DS}) is _____.

Q17. A single-phase transformer is designed to operate at 2 V per turn and turns ratio of 3:1. If the secondary winding is to supply a load of 8 kVA at 80 V, then the primary voltage is _____, the number of turns on secondary winding is _____, the current in primary winding is _____.

Q18. A 220V DC shunt generator has an armature current of 10 A. The armature resistance of the generator is 0.5Ω . The value of the induced (generated) emf is _____ V.

Q19. A 220V DC shunt motor draws a line current of 15 A and its shunt field resistance is 110Ω . The value of armature current is _____ A.

Q20. A 4 pole, 50 Hz induction motor runs at 1410 rpm. The slip of this motor is _____.

Note: Attempt all the parts of a question in sequence.

Make neat solutions showing all the necessary steps

- Q.1 For the circuit shown in Figure Q1, determine the loop currents, i_1 , i_2 , i_3 and i_4 using mesh analysis. [12]

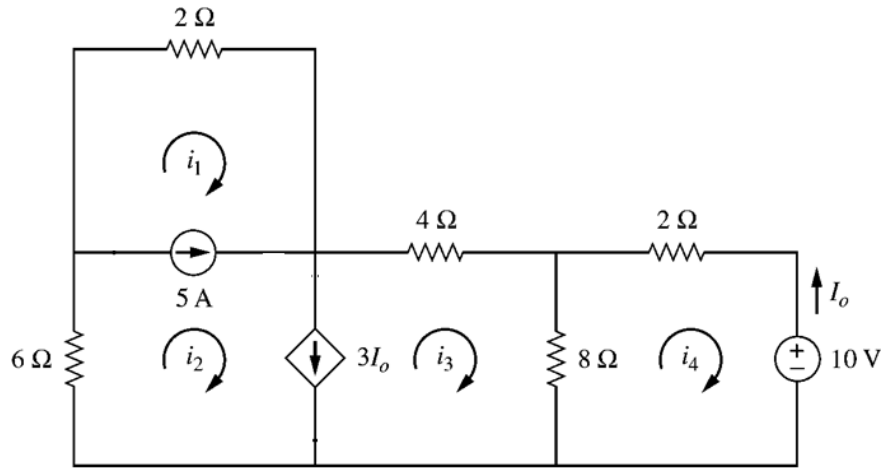


Figure Q1

- Q.2 A single phase source feeds a load having resistor and inductor in series. The average power drawn by the load is 200 kW at 0.707 pf lagging and the source voltage is $2000\sqrt{2} \cos 314t$.

- Determine the value of resistance and inductance of the load.
- Now, a capacitor is added in parallel to the load to improve the power factor to 0.85 lagging, calculate the value of this capacitance. [12]

- Q.3 Consider the circuit shown in Figure Q3 (a), the characteristics of the Zener diode is shown in Figure Q3 (b). For this circuit

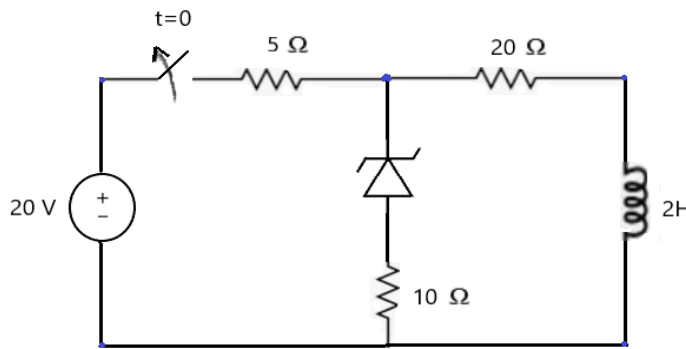


Figure Q 3(a)

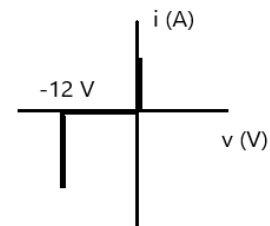


Figure Q 3(b)

- When the switch was closed for a long time ($t < 0$), determine whether the Zener diode operated in the forward/breakdown region.

- (ii) When the switch is opened at $t = 0$, the Zener diode works in forward biased. Obtain $i_L(t)$ and $v_L(t)$ for all times and also sketch their nature. [15]

Q.4 In the circuit shown in Figure Q4, the silicon pnp -BJT has $\alpha = 0.98$ and the silicon nnp -BJT has $\beta = 100$. $R_1 = R_2 = 1\text{ k}\Omega$, and $R_B = 60\text{ k}\Omega$. To start with, assume active region operation for both BJTs.

- (i) Determine: i_1 , i_2 , and i_3 .
(ii) Verify that Q_1 is operating in the active region.
(iii) Determine the maximum value of R_3 for Q_2 to operate in the active region. [16]

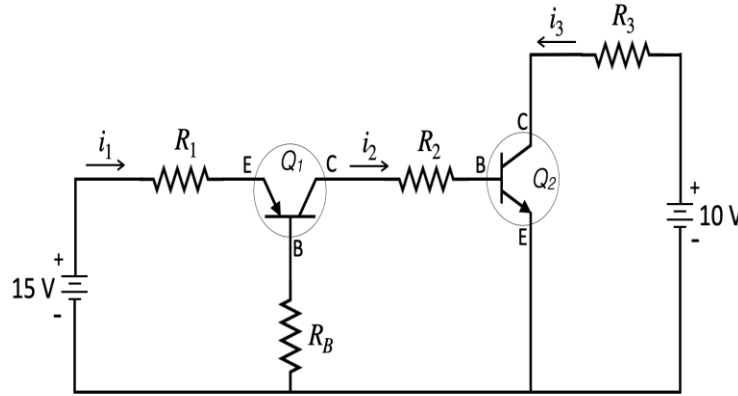


Figure Q4

Q.5 Both the JFETs in the circuit shown in Figure Q5 have $I_{DSS} = 16\text{ mA}$ and $V_p = -4\text{ V}$. Given that $R_S = 1\text{ k}\Omega$, $V_{DD} = 6\text{ V}$, the lower JFET is in the active region, and the upper JFET is in the ohmic region, find (a) i_D and (b) V_{DS} .

The current expressions in different region of operations is given below, choose the appropriate expressions. [20]

$$i_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_p} \right)^2$$

$$i_D = I_{DSS} \left[2 \left(1 - \frac{V_{GS}}{V_p} \right) \frac{V_{DS}}{-V_p} - \left(\frac{V_{DS}}{V_p} \right)^2 \right]$$

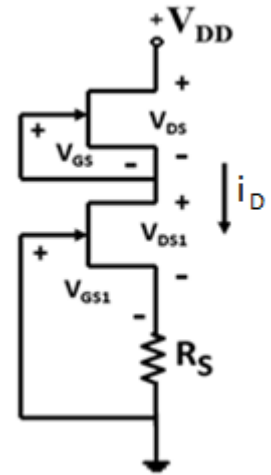


Figure Q5

Q6. The magnetically coupled circuit shown in Figure Q6 has a coupling coefficient of 1 ($k=1$). $L_1 = 2.653\text{ mH}$ and $L_2 = 10.61\text{ mH}$. Determine:

- (i) the loop currents i_1 and i_2
(ii) the total energy stored at time, $t = 5\text{ ms}$
(iii) the average power absorbed by $4\text{ }\Omega$ resistor. [15]

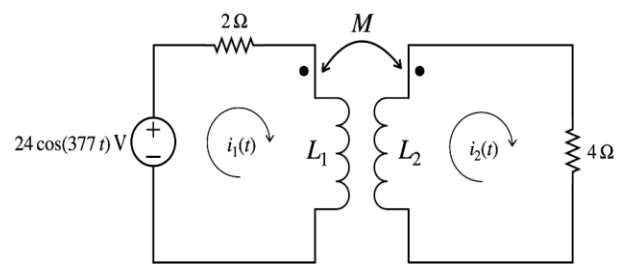


Figure Q6